

Total No. of Questions : 8]

SEAT No. :

PB-3846

[Total No. of Pages : 2

[6262]-108

T.E. (Electronics & Computer Engineering)

DATA COMMUNICATION

(2019 Pattern) (Semester - I) (310343)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) *Answers the Q.1 or Q.2 and Q.3 or Q.4 and Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of calculator is allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1) a)** Explain QPSK with its Transmitter. **[8]**
b) Explain M-ary FSK in detail with the help of Transmitter and Receiver. **[8]**

OR

- Q2) a)** For a given data (101101), draw the modulated output waveforms for **[8]**
i) BASK
ii) BFSK
iii) BPSK
iv) QPSK
b) Explain with the help of neat block diagram Explain Binary Frequency Shift Keying (BFSK) Transmitter and receiver. **[8]**

- Q3) a)** Explain different Multiple Access Techniques in detail. **[9]**
b) A PN sequence is generated using feedback shift registers of length $m = 4$, Find the generated output sequence if the initial contents of the shift register are 1000. If the chip rate is 10^7 chips/sec. Find the following Parameter **[9]**
i) Design PN sequence
ii) Chip Duration
iii) Length of PN sequence
iv) Duration of PN Sequence

OR

P.T.O.

- Q4) a)** With the help of mathematical expressions and block diagram explain DS-SS system. [9]
- b)** Write a short note on [9]
- Pure ALOHA
 - Slotted ALOHA
 - CSMA

- Q5) a)** Apply Shannon Fano coding procedure to find the coding efficiency for the following Message ensemble. $X_1 = 1/4$, $X_2 = 1/8$, $X_3 = 1/16$, $X_4 = 1/16$, $X_5 = 1/16$, $X_6 = 1/4$, $X_7 = 1/16$, $X_8 = 1/18$ Take $M = 2$. [9]
- b)** Define information rate, Entropy, Mutual information and channel capacity, Minimum Distance. [9]

OR

- Q6) a)** An information source produces a sequence of independent symbols having the following probabilities. $S_1 = 1/3$, $S_2 = 1/27$, $S_3 = 1/3$, $S_4 = 1/9$, $S_5 = 1/9$, $S_6 = 1/27$, $S_7 = 1/27$. Construct using Huffman binary encoding procedure and find its efficiency. [9]
- b)** State and Explain shannons theorem on channel capacity. [9]

- Q7) a)** Explain in detail ARQ Stop and Wait & Selective repeat ARQ. [6]
- b)** Explain the need of Error detection and Error Correction. [6]
- c)** What is Generator matrix and parity matrix? What is their role? [6]

OR

- Q8) a)** The Generator matrix of a particular (6, 3) block code is given below. Find all code vectors of this code [8]

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

- b)** Compare Forward error correction and Automatic repeat request. [4]
- c)** Write a short note on Linear block code. [6]



SEAT No. :

PA-3995

[Total No. of Pages : 3

[5926]-220

T.E. (Electronics and Computer Engineering)

DATA COMMUNICATION

(2019 Pattern) (Semester - I) (310343)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of calculator is allowed.

- Q1)**
- a) Explain QPSK signal generation with neat block diagram. [6]
 - b) With the help of block diagram and waveforms, explain generation of coherent BFSK. [6]
 - c) It is required to transmit 2.08×10^6 binary digit per second with $P_b \leq 10^{-6}$. Two possible schemes are considered. [8]
 - i) BPSK
 - ii) 16-Ary PSK

The channel noise PSD is $S_n(\omega) = 10^{-8}$. Determine the transmission bandwidth and the signal to power required at receiver input in each case.

Given : $\text{erf}(0.99999) = 3.3$

OR

- Q2) a)** Compare the performance of modulation schemes, BPSK, BFSK, QPSK, DPSK, M-ary PSK, M-ary FSK w.r.t. **[6]**
- i) Variable characteristics in o/p carrier
 - ii) BW
 - iii) Symbol duration
 - iv) Probability of Error.
 - v) Bit per symbols
 - vi) Applications
- b)** Explain generation. Detection of BPSK system with neat block diagram. **[6]**

P.T.O.

- c) Binary data is transmitted using M-ary PSK at a rate 2 Mbps over RF link having bandwidth 2MHz find signal power required at the receiver input so that bit error probability is less than 10^{-5} . [8]

Given : $M = 16$ and $M = 32$ and Noise PSD $\frac{N_0}{2} = 10^{-8}$ Watt / Hz,
 $\text{erf}(0.99995) = 3.2$.

- Q3) a)** State and explain properties of PN sequence. State the advantages of PN sequence. [8]
b) Explain the following frequency of spread spectrum systems with help of relevant diagrams. [8]
- Slow frequency hopping
 - Fast frequency hopping

OR

- Q4) a)** Draw the block diagram of DSSS BPSK transmitter and receiver system and explain various blocks. [8]
b) Differentiate between DHSS and FHSS w.r.t. [8]
- i) Definition
 - ii) Chip rate
 - iii) Modulation techniques
 - iv) Processing gain
 - v) Error probability
 - vi) Acquisition time
 - vii) Effect of distance
 - viii) Applications

- Q5) a)** Define Mutual Information? List properties of Mutual information? [8]
b) State & Explain [8]
- i) Shannons channel coding theorem
 - ii) Shannon's Information capacity theorem

OR

- Q6) a)** Compare Shannon - Fano - and Huffman coding techniques. [8]
b) A source emits 1000 symbols per second from a range of 5 symbols, with probabilities $\left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} \right\} \frac{1}{16}$ find source entropy and information rate. [8]

- Q7) a)** What is stop - and - wait ARQ? Explain. [6]
- b)** For a systematic LBC, the three parity check digits C4, C5 and C6 are given by [6]
- $$C4 = d1 \oplus d2 \oplus d3$$
- $$C5 = d1 \oplus d2$$
- $$C6 = d1 \oplus d3$$
- i) Construct generator matrix.
- ii) Construct code generated by this matrix.
- iii) Determine error correcting capability.
- c)** Define systematic, non-systematic code and Explain properties of linear block code. [6]

OR

- Q8) a)** Define the following : [6]
- i) Code rate
- ii) Word length
- iii) Hamming distance
- iv) Minimum hamming distance
- v) Block length
- vi) Constraint length
- b)** For a systematic LBC, the parity check bits are [6]
- $$C_1 = M_1 \oplus M_2 \oplus M_3$$
- $$C_2 = M_2 \oplus M_3 \oplus M_4$$
- $$C_3 = M_1 \oplus M_2 \oplus M_4$$
- Find
- i) Generator matrix.
- ii) Error detecting & correcting capabilities.
- iii) Parity Check Matrix.
- iv) Corrected code word for received code word [1101001].
- c)** Comment whether following code is perfect code or not, with necessary justification. [6]
- i) (7, 4) LBC
- ii) (6, 3) LBC



Total No. of Questions : 8]

SEAT No. :

P7605

[6180]-124

[Total No. of Pages : 3

T.E. (Electronics and Computer Engineering)

DATA COMMUNICATION

(2019 Pattern) (Semester - I) (310343)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of Calculator is allowed.

Q1) a) Explain QPSK signal generation with neat block diagram. **[6]**

b) With the help of block diagram and waveforms, explain generation of coherent BFSK. **[6]**

c) It is required to transmit 2.08×10^6 binary digit per second with $P_b \leq 10^{-6}$. Two possible schemes are considered. **[8]**

i) BPSK

ii) 16-Ary PSK

The channel noise PSD is $S_n(\omega) = 10^{-8}$. Determine the transmission bandwidth and the signal to power required at receiver input in each case. Given : $\text{erf}(0.99999) = 3.3$

OR

Q2) a) Compare the performance of modulation schemes, BPSK, BFSK, QPSK, DPSK, M-ary PSK, M-ary FSK w.r.t. **[6]**

i) Variable characteristics in o/p carrier.

ii) BW.

iii) Symbol duration

iv) Probability of Error.

v) Bit per symbols.

vi) Applications

P.T.O.

- b) Explain generation, Detection of BPSK system with neat block diagram. [6]
- c) Binary data is transmitted using M-ary PSK at a rate 2 Mbps over RF link having bandwidth 2MHz find signal power required at the receiver input so that bit error probability is less than 10^{-5} . [8]

Given: $M = 16$ and $M=32$ and Noise PSD $\frac{N_0}{2} = 10^{-8}$ Watt / Hz, erf (0.99995) = 3.2.

- Q3)** a) Explain different Multiple Access Techniques in detail. [8]
- b) A PN sequence is generated using feedback shift registers of length $m=4$, Find the generated output sequence if the initial contents of the shift register are 1000. If the chip rate is 10^7 chips/sec. Find the following Parameter [8]
- Design PN sequence
 - Chip Duration
 - Length of PN sequence
 - Duration of PN Sequence

OR

- Q4)** a) Draw the block diagram of DSSS BPSK transmitter and receiver system and explain various blocks. [8]
- b) Differentiate between DHSS and FHSS w.r.t. [8]
- Definition
 - Chip rate
 - Modulation techniques
 - Processing gain
 - Error probability
 - Acquisition time
 - Effect of distance
 - Applications

- Q5)** a) State & Explain [8]
- Shannon's channel coding theorem
 - Shannon's Information capacity theorem
- b) Define information rate, Entropy, Mutual information and channel capacity. [8]

OR

- Q6)** a) Compare Shannon - Fano - and Huffman coding techniques. [8]
 b) A source emits 1000 symbols per second from a range of 5 symbols, with probabilities. [8]

$\left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16} \right\}$ find source entropy and information rate.

- Q7)** a) What is stop - and - wait ARQ? Explain. [6]
 b) Explain the need of Error detection and Error Correction. [6]
 c) Define systematic, non-systematic code and Explain properties of linear block code [6]

OR

- Q8)** a) Define the following: [6]

- i) Code rate
- ii) Word length
- iii) Hamming distance
- iv) Minimum hamming distance
- v) Block length
- vi) Constraint length

- b) For a systematic LBC, the parity check bits are [6]

$$C_1 = M_1 \oplus M_2 \oplus M_3$$

$$C_2 = M_2 \oplus M_3 \oplus M_4$$

$$C_3 = M_1 \oplus M_2 \oplus M_4$$

Find

- i) Generator matrix.
 - ii) Error detecting & correcting capabilities.
 - iii) Parity Check Matrix.
 - iv) Corrected code word for received code word [1101001].
- c) Comment whether following code is perfect code or not, with necessary justification. [6]
- i) (7, 4) LBC
 - ii) (6,3) LBC

